# Mughees Ur Rehman

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## Education

**Virginia Tech** — **August 2024 – May 2026**
*Masters in Computer Science* — *Blacksburg, VA, USA*
**Cumulative GPA: 4.00/4.00**

**Lahore University of Management Sciences (LUMS)** — **September 2020 – June 2024**
*Bachelors in Computer Science* — *Lahore, Pakistan*
**Cumulative GPA: 3.97/4.00**

## Publications

**Edge Caching as Differentiation** — **September 2025**
*ACM SIGCOMM 2025* — *Coimbra, Portugal*
- Muhammad Abdullah, **Mughees Ur Rehman**, Pavlos Nikolopoulos, Katerina Argyraki.
- Paper
- Best Student Paper Award

**Source Code Hotspots: A Diagnostic Method for Quality Issues** — **April 2026**
*ACM MSR 2026* — *Rio de Janeiro, Brazil*
- Saleha Muzammil, **Mughees Ur Rehman**, Zoe Kotti, Diomidis Spinellis.
- Paper

**Towards Fairer AI: Multi-Agent Debiasing of LLMs With Online Evidence Retrieval** — **November 2025**
*AAAI Fall 2025 Symposium* — *Arlington, VA, USA*
- **Mughees Ur Rehman**, Saleha Muzammil.

## Research Experience

**Data Security and Privacy Lab, Virginia Tech** — **September 2024 – Present**
*Research Associate* — *Blacksburg, VA, USA*
- Conducting research with Prof. Murat Kantarcioglu on using LLMs to improve network security by automatically generating and evaluating Suricata intrusion detection rules.
- Designed **RulePilot**, an agentic AI system that uses **GPT-OSS-120B** and **GPT-5.5** as backbone models to generate, verify, and repair deployable Network Intrusion Detection System (NIDS) rules directly from raw malware network traffic.
- Constructed a malware traffic benchmark of **1,296** PCAPs across **192** malware families, correlating protocol-aware flows extracted with **tshark** against Suricata rules to create flow-level rule-generation ground truth.
- Developed benign traffic filtering and execution-grounded validation pipelines that removed **88.8%** of background flows, reduced LLM token cost by **1.99×**, and achieved **85.3%** rule syntax validity with near-zero benign false positives.

**EPFL** — **May 2024 – August 2024**
*Research Intern* — *Lausanne, Switzerland*
- Worked under the supervision of Prof. Katerina Argyraki at the Network Architecture Lab (NAL), investigating how edge caching affects Internet fairness and neutrality across major CDN providers.
- Developed Selenium-based crawlers in Python, deployed on AWS EC2 instances across multiple regions, to measure cache hit rates and latency across 25+ streaming services.
- Analyzed disparities in cache performance within CDNs and showed how such variations can lead to QoE differences and traffic prioritization among competing content providers.
- Co-authored the paper Edge Caching as Differentiation, published at **ACM SIGCOMM 2025**(Best Student Paper Award).

**Networks and Systems Group LUMS** — **June 2022 – May 2024**
*Research Associate* — *Lahore, Pakistan*
- Worked under the supervision of Prof. Zafar Ayyub Qazi at the Networks and Systems Group (NSG), focusing on edge computing and stateful application migration in 5G environments.
- Developed a low-latency key-value datastore based on a custom hashed queue data structure to enable efficient state migration on the network edge.(Project Repository)
- Initiated the project Re-thinking Redis for Edge Networks, addressing bottlenecks in Redis's blocking migration API and transforming it into an asynchronous, scalable operation.

## Professional Experience

**Analytics 4 Everyone** **June 2025 – August 2025**
*Software Engineering Intern* *Pittsburgh, PA*

- Developed full-stack features using React and Django, with PostgreSQL as the database backend, for an AI-powered education platform designed to scale for 10,000+ concurrent users and 1M+ total users.
- Created an automated data pipeline with task scheduling and GCP Bucket integration for scalable storage.

**Educative Inc.** **June 2023 – August 2023**
*Technical Content Engineering Intern* *Lahore, Pakistan*

- Authored 60+ technical articles on software engineering and applied mathematics, published on the Educative platform and optimized for discoverability through SEO best practices. (Educative Article Publications)
- Implemented Docker based workflows to support Educative's embedded code runner, enabling users to execute interactive code examples directly within the articles.

## Teaching Experience

**Virginia Tech** **August 2024 – May 2026**
*Graduate Teaching Assistant* *Blacksburg, VA, USA*

- CS 2114 Software Design & Data Structures , CS 5740 AI Tools for Software Engineering, CS 3114 Data Structures & Algorithms

**LUMS** **August 2022 – May 2024**
*Undergraduate Teaching Assistant* *Lahore, Pakistan*

- CS 582 Distributed Systems, CS 535 Machine Learning,CS 473 Network Security, CS 210 Discrete Mathematics, CS 100 Computational Problem Solving

## Honors & Awards

- Received the SIGCOMM 2025 Best Student Paper Award.
- Awarded $1,500 CCI Commonwealth Cyber Initiative Grant (2024–2025) to support conference travel and academic development, recognized as a Cyber Innovation Scholar.
- Recipient of a full scholarship to attend the Richard Tapia Conference 2025.
- Awarded the LUMS Merit Scholarship, granted to the top 15 students in the batch with the highest academic standing.
- Placed on the LUMS Dean's Honor List (2020–2024).

## Academic Service

### Artifact Evaluator

- ACM CoNEXT 2026
- USENIX Security 2026 (Details)
- ACM SIGCOMM 2025 (Details)

### Reviewer

- ACM Internet Measurement Conference (IMC '26), Shadow TPC
- ACM Computing Surveys (CSUR) 2025

## Conferences & Presentations

**AAAI 2025 Fall Symposium Series** **November 2025**
*Presenter* *Arlington, VA, USA*

- Presented the short paper "Towards Fairer AI: Multi-Agent Debiasing of LLMs with Online Evidence Retrieval".

**Tapia Conference 2025** **September 2025**
*Attendee* *Dallas, TX, USA*

- Participated in workshops, career fairs, and represented Virginia Tech at the graduate school fair.

## Technical Skills

**Languages**: Python, JavaScript, TypeScript, C++, C, C#, Go
**Cloud Infrastructure**: Google Cloud Platform, AWS, Azure
**Frontend Frameworks**: React, React Native, Angular
**Backend Frameworks**: Node.js, Django, FastAPI, Flask, Spark
**Databases**: MySQL, MongoDB, Postgres, Firebase
**DevOps (CI/CD)**: Git, Docker, Kubernetes, Jira